

Commit 6e25aff

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ActuallyNull committed on Aug 10

made a VERY loose model architecture i saw from a research paper and just thought i should document it

main 1 parent c9b2738 commit 6e25aff

Filter files...

data_preperation.py

model.py

2 files changed +62 -8 lines changed

Search within code

data_preperation.py +14 -8

@@ -4,14 +4,20 @@

4 4 from sklearn.model_selection import train_test_split

5 5

6 6

7 - X = []

8 - y = []

9 7

10 - X_processed, y_processed = preprocess_dataset("training2017/training2017", "training2017/training2017/REFERENCE.csv", X,

11 y)

12 - X_cnn = X_processed.reshape((X_processed.shape[0], X_processed.shape[1], 1))

13 - le = LabelEncoder()

14 - y_cnn = le.fit_transform(y_processed)

9 + def create_train_val_test_splits():

15 10

16 - X_train, X_temp, y_train, y_temp = train_test_split(X_cnn, y_cnn, test_size=0.3, random_state=42)

17 - X_val, X_test, y_val, y_test = train_test_split(X_temp, y_temp, test_size=0.5, random_state=42)

11 + X = []

12 + y = []

13 +

14 + X_processed, y_processed = preprocess_dataset("training2017/training2017", "training2017/training2017/REFERENCE.csv",

15 X, y)

16 + X_cnn = X_processed.reshape((X_processed.shape[0], X_processed.shape[1], 1))

17 + le = LabelEncoder()

18 + y_cnn = le.fit_transform(y_processed)

19 +

20 + X_train, X_temp, y_train, y_temp = train_test_split(X_cnn, y_cnn, test_size=0.3, random_state=42)

21 + X_val, X_test, y_val, y_test = train_test_split(X_temp, y_temp, test_size=0.5, random_state=42)

22 +

23 + return X_train, y_train, X_val, y_val, X_test, y_test

model.py +48

@@ -0,0 +1,48 @@

1 + import tensorflow as tf

2 + from keras import layers

3 + from data_preperation import create_train_val_test_splits

4 + from sklearn.metrics import confusion_matrix, classification_report

5 +

6 + X_train, y_train, X_val, y_val, X_test, y_test = create_train_val_test_splits()

7 +

8 + model = tf.keras.Sequential([

9 + layers.Conv1D(32, kernel_size=5),

10 + layers.Conv1D(32, kernel_size=5),

11 + layers.MaxPool1D(pool_size=2),

12 + layers.Conv1D(64, kernel_size=5),

13 + layers.Conv1D(64, kernel_size=5),

14 + layers.MaxPool1D(pool_size=2),

15 + layers.Conv1D(128, kernel_size=5),

16 + layers.Conv1D(128, kernel_size=5),

17 + layers.Conv1D(128, kernel_size=5),

18 + layers.MaxPool1D(pool_size=2),

19 + layers.Conv1D(256, kernel_size=5),

20 + layers.Conv1D(256, kernel_size=5),

21 + layers.Conv1D(256, kernel_size=5),

22 + layers.MaxPool1D(pool_size=2),

23 + layers.Conv1D(256, kernel_size=5),

24 + layers.Conv1D(256, kernel_size=5),

25 + layers.Conv1D(256, kernel_size=5),

26 + layers.MaxPool1D(pool_size=2),

27 + layers.GlobalAveragePooling1D(),

28 + layers.Dense(128, activation='relu'),

29 + layers.Dense(3, activation='softmax')

30 +])

31 +

32 + model.compile(optimizer='adam',

33 + loss='sparse_categorical_crossentropy',

34 + metrics=['accuracy'])

35 +

36 + model.fit(X_train, y_train, epochs=10, batch_size=32)

37 +

38 + test_loss, test_acc = model.evaluate(X_test, y_test)

39 + print(f"Test accuracy: {test_acc:.4f}")

40 +

41 + y_pred = model.predict(X_test)

42 + y_pred_classes = y_pred.argmax(axis=1)

43 +

44 + print("Confusion Matrix:")

45 + print(confusion_matrix(y_test, y_pred_classes))

46 +

47 + print("Classification Report:")

48 + print(classification_report(y_test, y_pred_classes))

Comments 0

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